Data Analysis & Probability

ANSWER KEY

Section 1: Mean, Median, Mode & Range

1 The test scores of five students are $75, 80, 85, 90, 95$. What is the mean score?

a. 85 ✓

b. 80

c. 90

d. 87

Mean $= \frac{75+80+85+90+95}{5} = \frac{425}{5} = 85$.

2 What is the median of the data set: $14, 8, 22, 11, 19, 5, 17$?

a. 14 ✓

b. 11

c. 17

d. 13

Order the numbers: $5, 8, 11, 14, 17, 19, 22$. The middle value (4th of 7) is 14.

3 The average of 4 numbers is 15. If a fifth number is added, the new average becomes 18. What is the fifth number?

a. 30 ✓

b. 25

c. 28

d. 32

Original sum $= 4 \times 15 = 60$. New sum $= 5 \times 18 = 90$. The fifth number $= 90 - 60 = 30$.

Section 2: Basic & Conditional Probability

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4 A fair six-sided die is rolled. What is the probability of obtaining an even number or a number greater than 4?

a. $\frac{2}{3}$ ✓

b. $\frac{1}{2}$

c. $\frac{5}{6}$

d. $\frac{1}{3}$

Even numbers: $\{2, 4, 6\}$. Numbers > 4: $\{5, 6\}$. Union: $\{2, 4, 5, 6\}$ (4 outcomes). Probability $= \frac{4}{6} = \frac{2}{3}$.

5 A bag contains 4 red marbles, 6 blue marbles, and 5 green marbles. If one marble is drawn at random, what is the probability that it is NOT blue?

a. $\frac{3}{5}$ ✓

b. $\frac{2}{5}$

c. $\frac{1}{3}$

d. $\frac{2}{3}$

Total marbles $= 4 + 6 + 5 = 15$. Non-blue marbles $= 4 + 5 = 9$. Probability $= \frac{9}{15} = \frac{3}{5}$.

6 Two coins are tossed simultaneously. What is the probability of getting at least one Head?

a. $\frac{3}{4}$ ✓

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{2}{3}$

Sample space: $\{HH, HT, TH, TT\}$ (4 outcomes). At least one head: $\{HH, HT, TH\}$ (3 outcomes). Probability $= \frac{3}{4}$.

Section 3: Circular Pie Charts & Sector Percentages

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7 In a circle graph representing a family's budget, the angle representing housing expenses is 90°. What percentage of the budget is spent on housing?

a. 25% ✓

b. 20%

c. 30%

d. 35%

A full circle is 360°. Housing percentage $= \frac{90^\circ}{360^\circ} \times 100\% = \frac{1}{4} \times 100\% = 25\%$.

8 A circle graph represents a company's sales of $720,000$ SAR. If the food sector has a central angle of 60°, what is the sales value of the food sector?

a. $120,000$ **SAR** ✓

b. $100,000$ SAR

c. $140,000$ SAR

d. $90,000$ SAR

Fraction $= \frac{60^\circ}{360^\circ} = \frac{1}{6}$. Sector sales $= \frac{1}{6} \times 720,000 = 120,000$ SAR.

9 If a sector of a circle graph represents 40% of the total, what is its central angle?

a. 144° ✓

b. 120°

c. 160°

d. 150°

Central angle $= 0.40 \times 360^\circ = 144^\circ$.

Section 4: Fundamental Counting Principle & Combinatorics

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10 A restaurant menu offers 4 appetizers, 5 main courses, and 3 desserts. How many different 3-course meals can a customer choose?

a. 60 ✓

b. 50

c. 45

d. 75

By the fundamental counting principle: $4 \times 5 \times 3 = 60$ distinct meals.

11 How many 3-digit numbers can be formed using the digits $\{1, 2, 3, 4, 5\}$ if repetition of digits is NOT allowed?

a. 60 ✓

b. 125

c. 20

d. 120

Hundreds digit: 5 choices. Tens digit: 4 choices. Units digit: 3 choices. Total numbers $= 5 \times 4 \times 3 = 60$.

12 In a meeting of 6 people, if everyone shakes hands with everyone else exactly once, how many handshakes take place?

a. 15 ✓

b. 30

c. 12

d. 20

Number of handshakes $= \frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15$.

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